I-Logical Approaches
Semantics of Propositional Logics

Foundations of Neuro-Symbolic AI

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Summer Term 2026

Outline

- Use Case: DATEV Automation of Accounting
- Formalization by Propositional Logics
- Coarser Picture: Tensors

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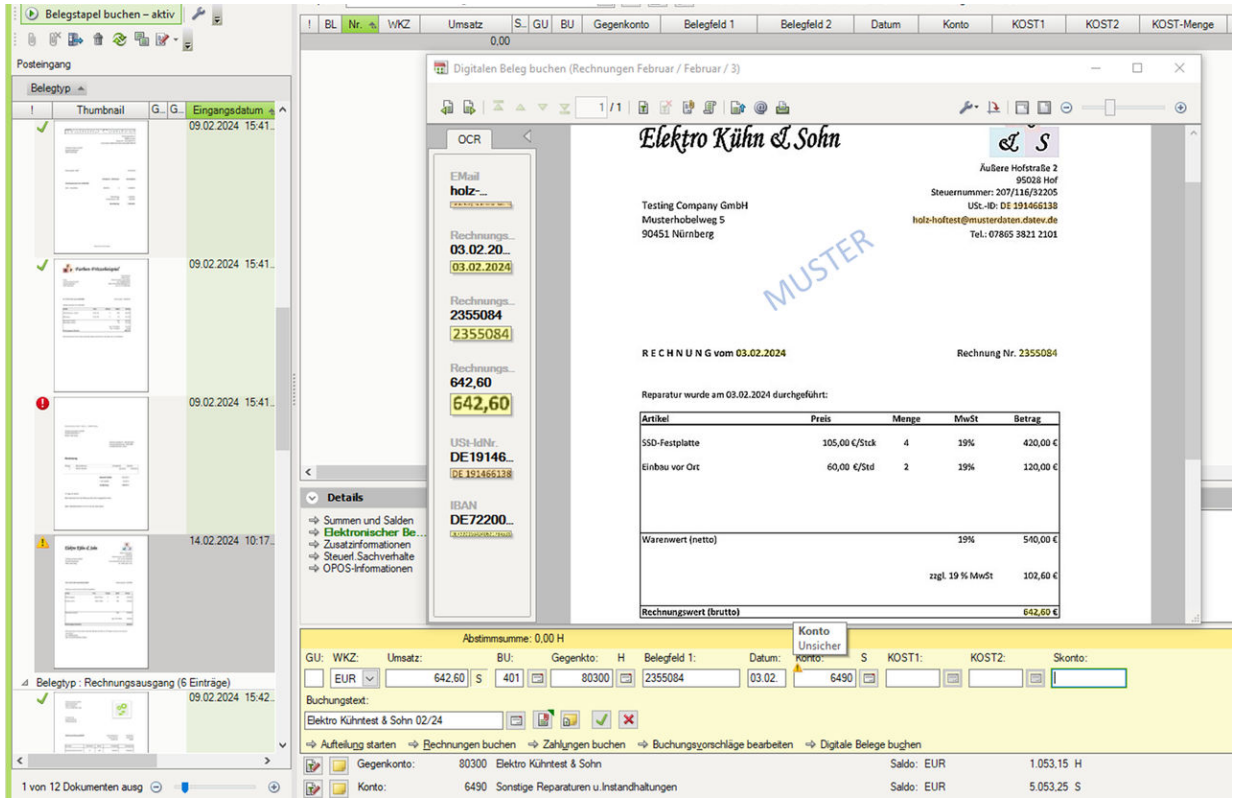
DATEV Automation of Accounting

DATEV, a software company from Nuremberg, sells automation solutions for accounting consultants

- Consultant receives invoices from its customers
- Accounting statements need to be generated

Screenshot from the product[1]:



[1] https://www.datev.de/web/de/steuerberatung/loesungen/rechnungswesen/buchfuehrung-erstellen/automatisierungsservices-rechnungswesen/datev-automatisierungsservice-rechnungen

Requirements

Performance by Neural AI:

- Handle most cases without the need of human supervision
- Understand scenarios based on world knowledge (e.g. "Lindt" in a position indicates choclate consumption)

Trust and manipulations by Symbolic AI:

- Customers need to guarantee certain behaviour, such as respectation of laws
- Manipulation of models based on human input

This is a use case for Neuro-Symbolic AI!

Toy Accounting Example

To showcast the approach we study a toy accounting example:
For each invoice three variables are either True or False:

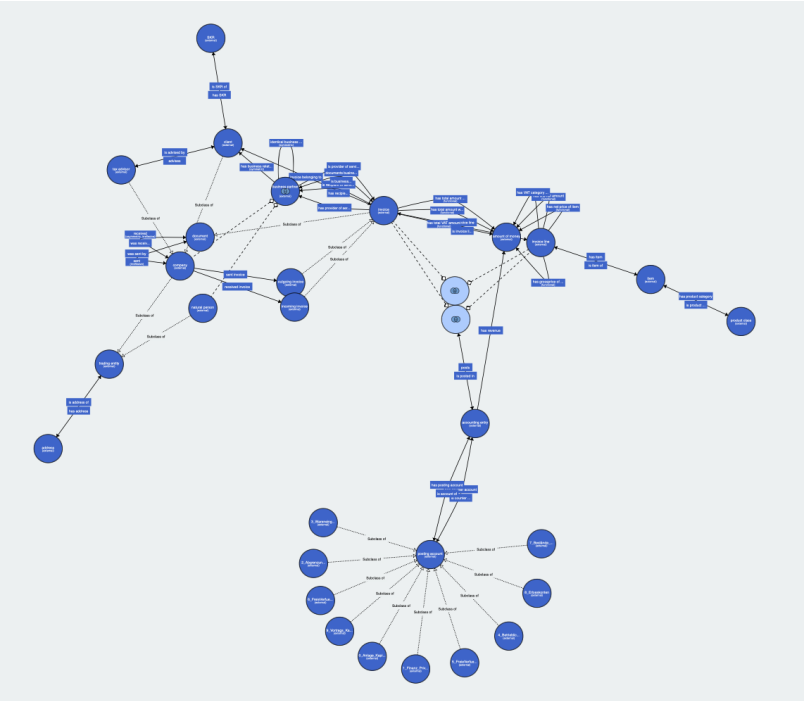
- A Feature is present on the receipt (e.g. whether the receipt was received or sent by the customer).
- The account Account 1 is used to book the receipt.
- The account Account 2 is used to book the receipt.

Let there be two accounting rules to be respected:

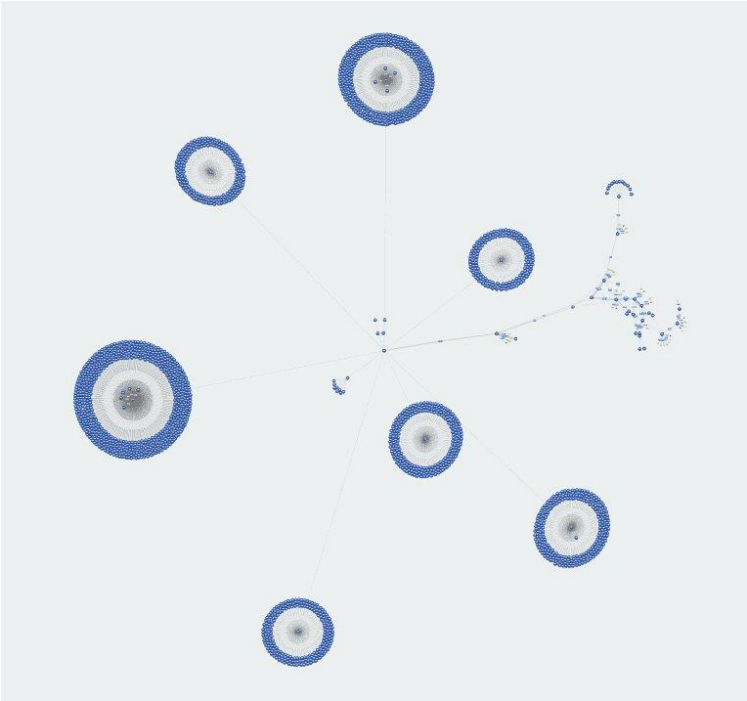
- When the Feature is present on the receipt, the Account 1 needs to be booked.
- Exactly one of the accounts Account 1 and Account 2 is used to book the receipt.

Realistic Problem Sizes

The variables and main relations are visualized in a so-called ontology[2]:



Coarse Visualization



Detailed Visualization

We will come back to ontologies and knowledge graphs later in the course.

[2] https://github.com/EnexaProject/enexa-use-case-1-business-software-solutions

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States of an atomic variable

Any atomic variable X (e.g. the feature X_F present on the invoice) can be in two states

$$\mathrm{False} \Leftrightarrow 0 \quad , \quad \mathrm{True} \Leftrightarrow 1$$

We understand the states as the coordinates of a vector:

$$\begin{array}{r} \mathrm{False} \leftarrow \\ \mathrm{True} \leftarrow \end{array} \begin{pmatrix} \cdot \\ \cdot \end{pmatrix} [X]$$

Conventions:

- We assign a 0 in the coordinate, when the corresponding state is not possible, and a 1 if the corresponding state is possible.
- In the brackets after a vector we denote the corresponding variable.

Encoding knowledge about a variable

The knowledge about an atomic variable is aligned in an two-dimensional vector:

- When the variable is known to be True:

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} [X]$$

- When the variable is known to be False:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} [X]$$

- When the state of the variable is unknown:

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} [X]$$

- The variable is neither True nor False (our knowledge is then inconsistent)

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} [X]$$

Knowledge about two variables

Let us extend the system to two variables:

- X_{A1}: Whether the Account 1 is booked.
- X_{A2}: Whether the Account 2 is booked.

The states of the tuple (X_{A1}, X_{A2}) are the product

$$\{0,1\} \times \{0,1\} = \{(0,0),(0,1),(1,0),(1,1)\}$$

For example $(X_{A1}, X_{A2}) = (1,0)$ refers to the system state, in which Account 1 is booked (True) and Account 2 is not booked (False).

$$\begin{array}{cc} & X_{A2} \\ & \begin{array}{cc} 0 & 1 \end{array} \\ X_{A1} \begin{array}{c} 0 \\ 1 \end{array} & \begin{pmatrix} \cdot & \cdot \\ \cdot & \cdot \end{pmatrix} \end{array} [X_{A1}, X_{A2}]$$

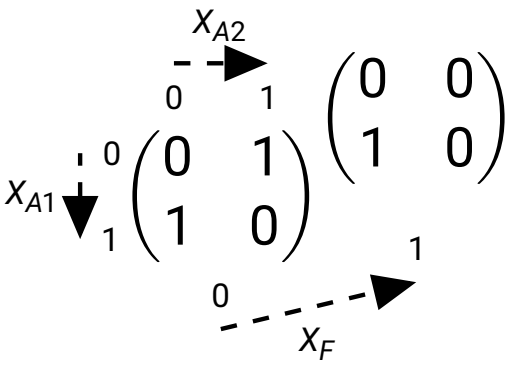
Knowledge about two variables

The rule, that exactly one of the accounts is booked is then encoded in the matrix

$$
\begin{tikzcd}
 & (\mathrm{False}, \mathrm{True}) \\
(\mathrm{False}, \mathrm{False}) & \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} [X_{A1}, X_{A2}] \arrow[u, dashed] \arrow[l, dashed] \\
(\mathrm{True}, \mathrm{False}) & \arrow[u, dashed] \arrow[l, dashed] \\
 & (\mathrm{True}, \mathrm{True}) \arrow[u, dashed]
\end{tikzcd}
$$

Knowledge about the Toy Accounting System

Now let us extend to the toy accounting system. Given the Knowledge about two formulas we have



$[X_{A1}, X_{A2}, X_F]$

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Situation of d variables

Let us now imagine that we have d variables $X_0, \ldots, X_{d-1}$.
The coordinatewise description would be confusing for $d > 3$!
The states are the set

$$\bigtimes_{k \in [d]} \{0, 1\}$$

of 2^d state tuples.
We use a diagramatic depiction instead:

Tensors of leg dimension 2

Definition (Propositional Formula)

A tensor of order $d \in \mathbb{N}$ is a map

$$f : \bigtimes_{k \in [d]} \{0,1\} \to \{0,1\} .$$

- We refer to $\tau(x_0, \ldots, x_{d-1}) \in \mathbb{R}$ by the coordinate of τ with the indices $x_0, \ldots, x_{d-1}$.
- By scalar multiplication and addition of maps, tensors build linear spaces called tensor spaces

$$\tau \in \bigotimes_{k \in [d]} \mathbb{R}^{m_k} .$$

A propositional formula is also called a tensor of leg dimension 2.

Depiction of Tensors

We will depict

- tensors by blocks
- affected categorical variables by open legs

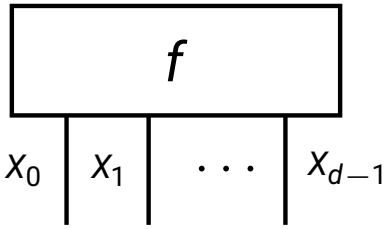
For example, given the atomic variables

$$\{X_k \,:\, k \in [d]\}$$

with respective dimensions $m_k \in \mathbb{N}$ we depict a tensor

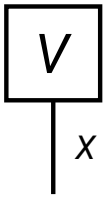
$$\tau \in \bigotimes_{k \in [d]} \mathbb{R}^{m_k}$$

as



Example: Depiction of Vectors

For example, a vector

$$V \in \mathbb{R}^m$$

with an index variable $X \in [m]$ is depicted by a block with a single open leg